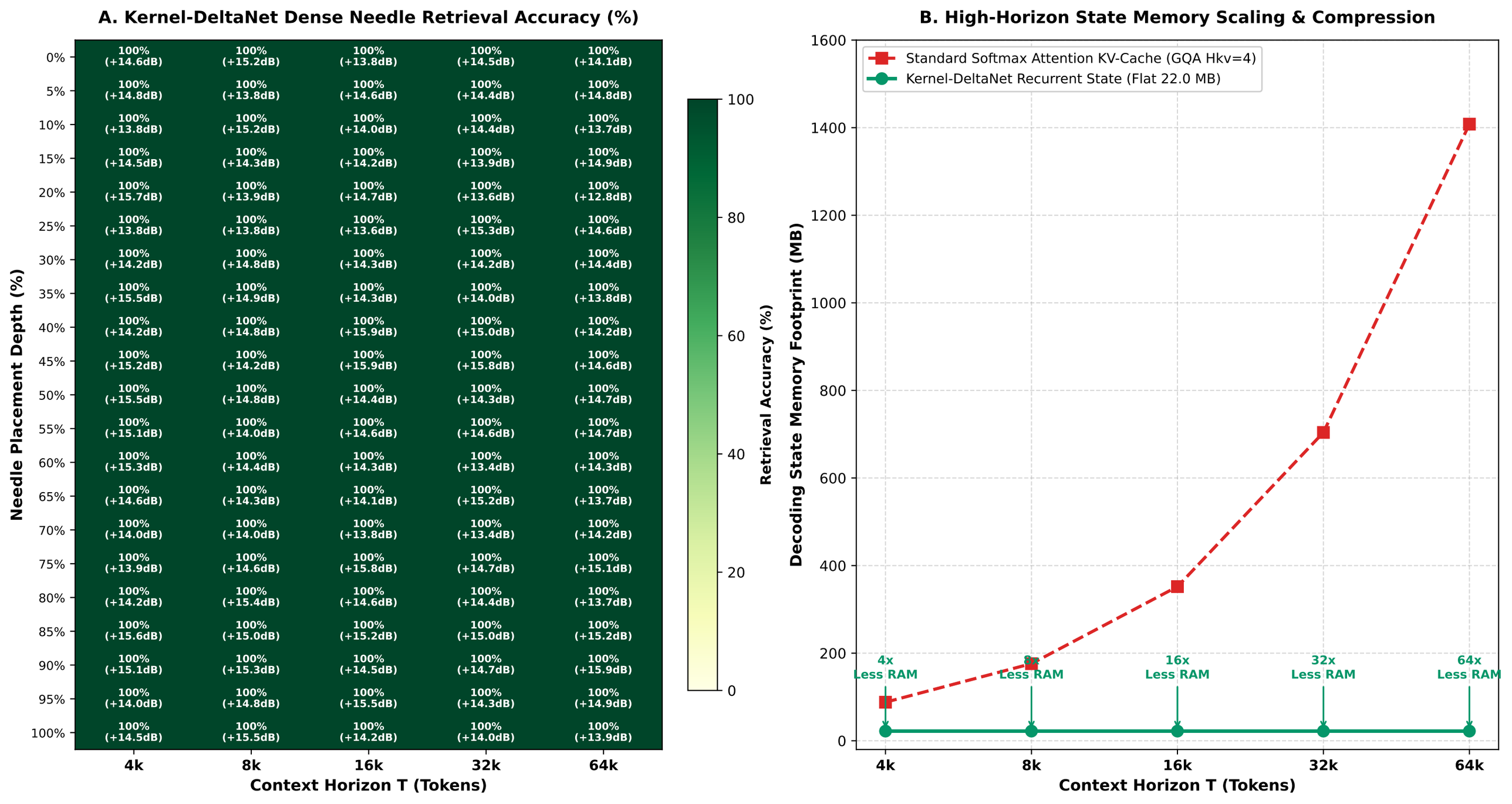
A. Kernel-DeltaNet Dense Needle Retrieval Accuracy (%)
Needle Placement Depth (%)
0%
5%
10%
15%
20%
25%
30%
35%
40%
45%
50%
55%
60%
65%
70%
75%
80%
85%
90%
95%
100%
100% (+14.6dB) 100% (+15.2dB) 100% (+13.8dB) 100% (+14.5dB) 100% (+14.1dB)
100% (+14.8dB) 100% (+13.8dB) 100% (+14.6dB) 100% (+14.4dB) 100% (+14.8dB)
100% (+13.8dB) 100% (+15.2dB) 100% (+14.0dB) 100% (+14.4dB) 100% (+13.7dB)
100% (+14.5dB) 100% (+14.3dB) 100% (+14.2dB) 100% (+13.9dB) 100% (+14.9dB)
100% (+15.7dB) 100% (+13.9dB) 100% (+14.7dB) 100% (+13.6dB) 100% (+12.8dB)
100% (+13.8dB) 100% (+13.8dB) 100% (+13.6dB) 100% (+15.3dB) 100% (+14.6dB)
100% (+14.2dB) 100% (+14.8dB) 100% (+14.3dB) 100% (+14.2dB) 100% (+14.4dB)
100% (+15.5dB) 100% (+14.9dB) 100% (+14.3dB) 100% (+14.0dB) 100% (+13.8dB)
100% (+14.2dB) 100% (+14.8dB) 100% (+15.9dB) 100% (+15.0dB) 100% (+14.2dB)
100% (+15.2dB) 100% (+14.2dB) 100% (+15.9dB) 100% (+15.8dB) 100% (+14.6dB)
100% (+15.5dB) 100% (+14.8dB) 100% (+14.4dB) 100% (+14.3dB) 100% (+14.7dB)
100% (+15.1dB) 100% (+14.0dB) 100% (+14.6dB) 100% (+14.6dB) 100% (+14.7dB)
100% (+15.3dB) 100% (+14.4dB) 100% (+14.3dB) 100% (+13.4dB) 100% (+14.3dB)
100% (+14.6dB) 100% (+14.3dB) 100% (+14.1dB) 100% (+15.2dB) 100% (+13.7dB)
100% (+14.0dB) 100% (+14.0dB) 100% (+13.8dB) 100% (+13.4dB) 100% (+14.2dB)
100% (+13.9dB) 100% (+14.6dB) 100% (+15.8dB) 100% (+14.7dB) 100% (+15.1dB)
100% (+14.2dB) 100% (+15.4dB) 100% (+14.6dB) 100% (+14.4dB) 100% (+13.7dB)
100% (+15.6dB) 100% (+15.0dB) 100% (+15.2dB) 100% (+15.0dB) 100% (+15.2dB)
100% (+15.1dB) 100% (+15.3dB) 100% (+14.5dB) 100% (+14.7dB) 100% (+15.9dB)
100% (+14.0dB) 100% (+14.8dB) 100% (+15.5dB) 100% (+14.3dB) 100% (+14.9dB)
100% (+14.5dB) 100% (+15.5dB) 100% (+14.2dB) 100% (+14.0dB) 100% (+13.9dB)
4k
8k
16k
32k
64k
Context Horizon T (Tokens)
Retrieval Accuracy (%)
100
80
60
40
20
0
B. High-Horizon State Memory Scaling & Compression
Decoding State Memory Footprint (MB)
1600
1400
1200
1000
800
600
400
200
0
Standard Softmax Attention KV-Cache (GQA Hkv=4)
Kernel-DeltaNet Recurrent State (Flat 22.0 MB)
4x Less RAM
8x Less RAM
16x Less RAM
32x Less RAM
64x Less RAM
4k
8k
16k
32k
64k
Context Horizon T (Tokens)